

Battery Warm-up Methodologies at Subzero Temperatures for Automotive Applications: Recent Advances and Perspectives (2019.11)

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Recently, Professor Xiaosong Hu of VPSL at Chongqing University and his collaborators (from Oxford, Berkeley, Queen Belfast and University of Maryland) were invited by Professor Hai Wang at Stanford University in the United States and Professor Christof Schulz at Duisburg-Essen University in Germany to publish a review article “Battery Warm-up Methodologies at Subzero Temperatures for Automotive Applications: Recent Advances and Perspectives” on the international top journal *Progress in Energy and Combustion Science* (Impact factor: 26.467). The effects of low temperature on the lithium-ion battery of electric vehicles and the thermal issues involved in the low-temperature warm-up of power batteries are summarized at the macro and micro scales. The design factors of low-temperature warm-up systems from the level of battery cells, battery modules/pack, and power systems are presented. This review also systematically elucidate all possible low-temperature preheating methods of power batteries at present and their latest research progress. The advantages and difficulties of different preheating methods, possible research trends of battery low-temperature thermal management from different perspectives are also discussed.

Article Overview

Research purpose

In order to promote sustainable energy development and respond to increasing energy demand, the development of electric vehicles has received widespread attention from around the world. As one of the core technologies of electric vehicles, the performance of power batteries directly affects the performance, safety, and economy of electric vehicles. However, the low-temperature problem has become a key technical difficulty in the development of global electric vehicles, which severely restricts the development of electric vehicles. At low temperatures, the discharge capacity and power of lithium-ion power batteries have been greatly reduced, and the internal resistance has increased sharply, which has caused problems such as shortened driving mileage, insufficient power, and reduced charging acceptance. In addition, lithium plating at the negative electrode is more significant at low temperatures, and the growth of lithium dendrites shortens the battery life and brings potential safety hazards. Although the low-temperature performance of lithium-ion batteries can be improved to some extent by improving battery materials, it takes a long time to ensure its overall performance over the full temperature range (−20 °C ~ 60 °C). The battery thermal management system (BTMS), by pre-heating the battery before operations to improve the performance of the power battery, has become another option to break through the low-temperature technology bottleneck of the power battery. Therefore, this article will provide strong support for thermal management and control of automotive power batteries in low-temperature environments.

Effects of low temperatures on Li-ion batteries

At the macro level, that is, the battery cell level, the impact of low temperature on lithium-ion batteries is characterized by the decrease of discharge capacity and available energy, increase of internal resistance and external output power. In addition, at low temperatures, the battery's charge acceptance is greatly reduced, and high-rate charging will also cause lithium plating at the negative electrode, causing irreversible loss of battery capacity. Therefore, in order to avoid lithium plating, the charging rate should be limited to a low level, which results in a significant increase in charging time. These cell-level performance degradations can be explained by micro-scale changes in battery material properties. At the micro-scale, the effect of low temperature on the battery negative electrode, positive electrode, and electrolyte materials is shown in Figure 1, which mainly includes: (1) the interface phenomenon of the negative electrode, such as the decrease in Li^+ permeability of the SEI film on the surface of graphite particles; (2) Slow kinetics in the electrode, such as increase in charge transfer resistance (R_{ct}) and intergranular resistance (R_{ct} at $-20\text{ }^\circ\text{C}$ can be 300% higher than room temperature value), and a decrease in the solid-phase Li^+ diffusion rate; 3) The increase of electrolyte viscosity and decrease of conductivity.

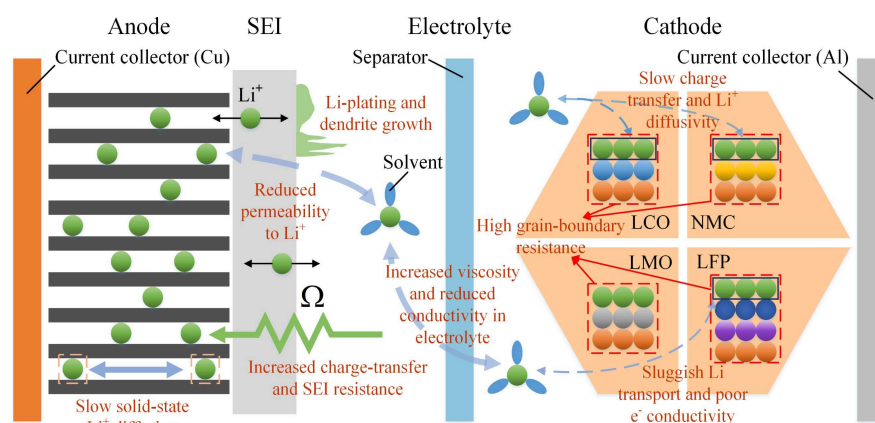


Fig. 1 Battery performance degradation mechanisms at low temperatures from a material perspective.

Thermal science involved in battery warm-up processes

Since the improvement of battery materials cannot enhance the overall performance of the power battery in the full temperature range in a short time, thermal management is adopted for the battery module/pack level, that is, the temperature of the battery pack is controlled within a suitable range by preheating to countervail the negative effects caused by low temperature, which is currently the most effective way to solve the problems brought by low temperature. In this way, improving the low-temperature performance of the battery becomes a thermal issue. In order to effectively preheat the power battery at low temperatures, it is necessary to fully understand the thermal performance of the battery and the related heat transfer characteristics related to the battery cell, module, and pack. At the same time, the heat generation, heat conduction or heat convection also need to be accurately modeled, measured and controlled in order to effectively regulate the temperature rise of the battery. Normally, the low-temperature preheating system of the power battery needs to consume a certain amount of energy. Therefore, it is necessary to comprehensively design the heat transfer method and path in the preheating system based on thermal theories and methods to reduce the heat loss during the preheating process, thereby reducing the energy consumption during the preheating process. At low temperatures, the main principle of the preheating process is shown in Figure 2.

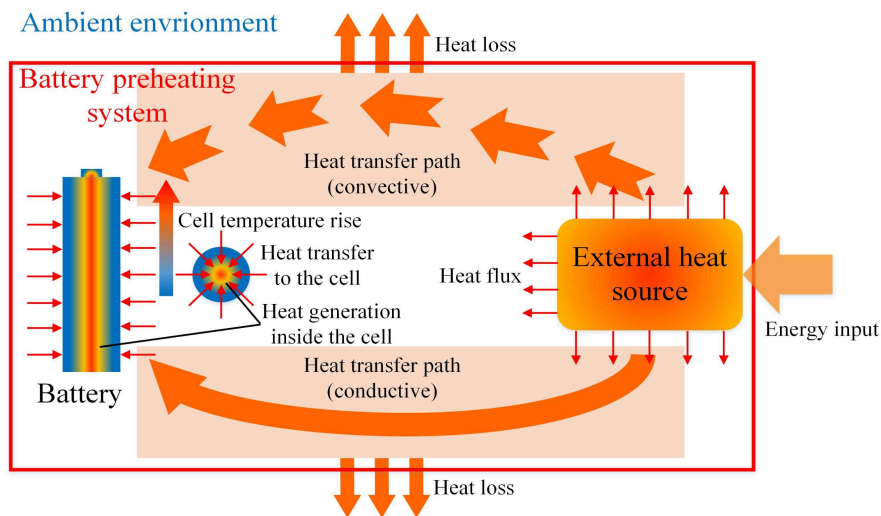


Fig. 2 Main principles of the preheating process.

Design consideration of the preheating system

The goal of battery thermal management at low temperatures is to restore the energy and power capability of Li-ion batteries as well as eliminate lithium plating through heating the batteries from a subzero temperature before the operation. In order to achieve fast warm-up while avoiding accelerated battery aging, the target temperature for warm-up is usually set to 0 °C, at which the battery can recover most of the available energy and power. For fast charging with charge rate higher than 1C, the target temperature of warm-up needs to be much higher than 0 °C to prevent lithium plating under such a high current rate. Therefore, the factors that need to be considered for low-temperature preheating of power batteries are summarized as follows: (a) The electrical energy consumption; (b) The preheating duration; (c) The effect of heating on battery aging; (d) The temperature uniformity of the battery cell, module, and pack; (e) The overall cost including the cost for the system, operational cost, and maintenance cost; (f) The complexity addition, such as the extra devices, weight, and space needed due to the integration of the heating system; (g) The safety and reliability of the heating system. In addition, for different battery system integration levels (battery cell, module/pack, system), the design considerations are also different, as shown in Figure 3 below:

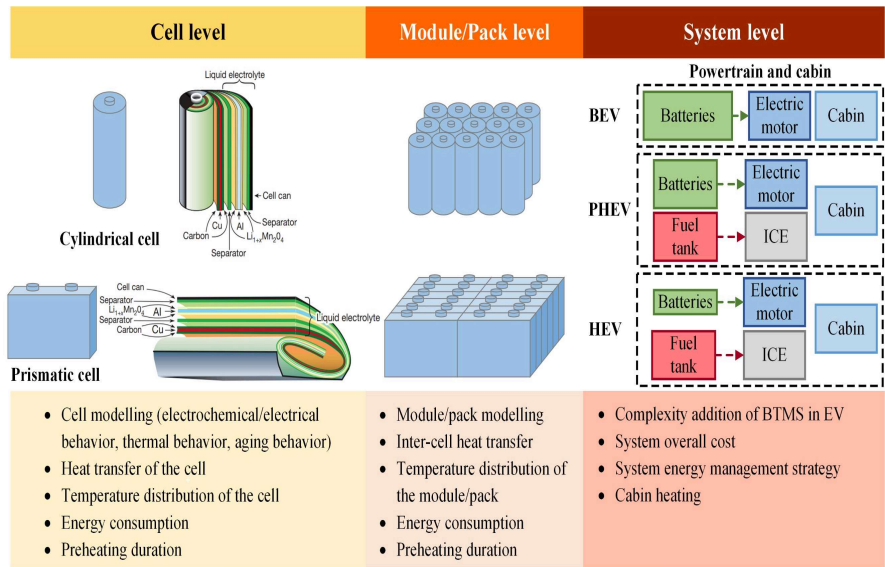


Fig. 3 Design considerations of BTMS at low temperatures at different battery integration levels. (Adapted from Tarascon JM, Armand M. Issues and challengesfacing rechargeable lithium batteries. Nature. 2001; 414: 359-67.)

Classification of battery warm-up strategies

The preheating strategies of power batteries can be classified according to the position of the heat source. Based on the boundary of the battery cell, the preheating methods can be mainly divided into two categories, as shown in Figure 4, including external heating and internal preheating. In external heating, the heat required for preheating is mainly provided by external heat sources (resistance wires or other electric heating elements), and the heat is then transferred to the power battery through convection or conduction. This process can be powered by either the battery or an external power supply (charging piles). At present, most commercial vehicles preheat batteries using external methods. Internal preheating is usually achieved by discharging or applying an alternating current to the battery. Using the internal resistance of the battery to generate heat to warm up the battery avoids the complicated heat transfer process and heat loss during the heat transfer process. A special internal preheating method, such as a self-heating lithium-ion battery, can significantly increase the heat production rate inside the battery by changing the battery structure and adding heat generation elements (nickel foil) inside the battery. Currently, most research focuses on internal preheating.

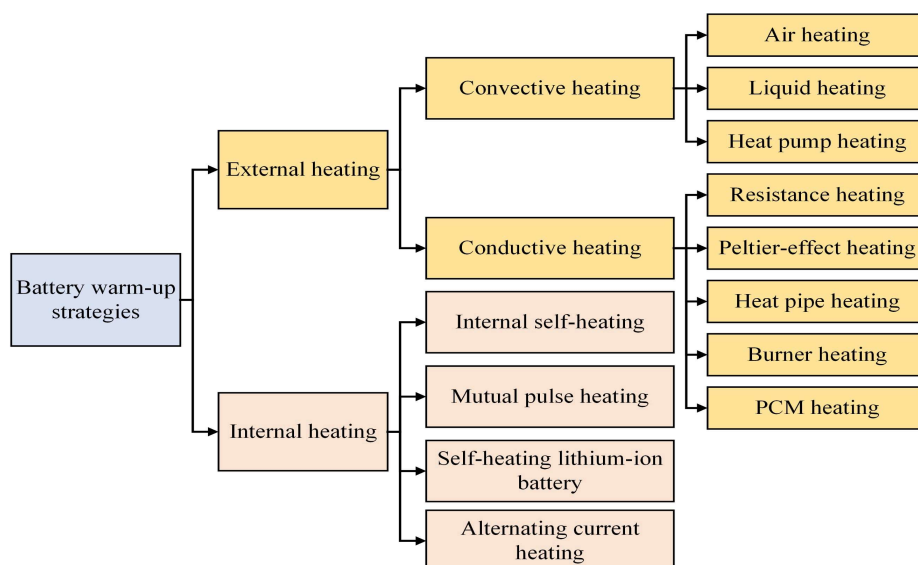


Fig. 4 Classification of preheating strategies.

External heating

Typically, external heating methods need specially designed BTMS to transfer the heat from external heat sources to battery cells during the preheating process, where factors such as the heat transfer patterns and routes, the geometry and layout of the battery cells, the properties of heat transfer media could affect the heating performance. Conventional BTMS, such as air and liquid systems, are economical for commercial applications with high safety and reliability. However, the low heat transfer rate and low heating efficiency are the main challenges in external heating, leading to a prolonged warm-up duration and increased energy consumption, especially when heating large-format Li-ion cells. Moreover, temperature difference within the battery cell and module/pack, affected by the cell geometry, the placement of heating elements, and the thermal conductivity of heat transfer medium, is another critical issue in conventional BTMS.

Efforts can be made in the future to solve the aforementioned problems by enhancing heating efficiency in external preheating systems. Specifically, in convective heating, increasing the heat transfer rate in BTMS is of great significance, which involves changing the flowing manner of heat transfer medium to achieve a higher heat transfer rate, applying novel heat transfer fluid with higher thermal conductivity (e.g. nanofluids), as well as optimizing the geometry of flow channels of the heat transfer medium. For conductive heating, better performance can be achieved through several methods: (a) optimizing the arrangement of battery cells to increase the effective contact area; (b) applying novel heating elements with higher

heating performance (e.g., film-based panel heaters) or the coefficient of performance; (c) increasing the thermal conductivity of the heat conduction medium such as the applications of nano-enhanced PCM. Besides these, better thermal insulation can also help to reduce the heat loss to the ambient environment during the heat transfer process, which can also reduce energy expenses during warm-up.

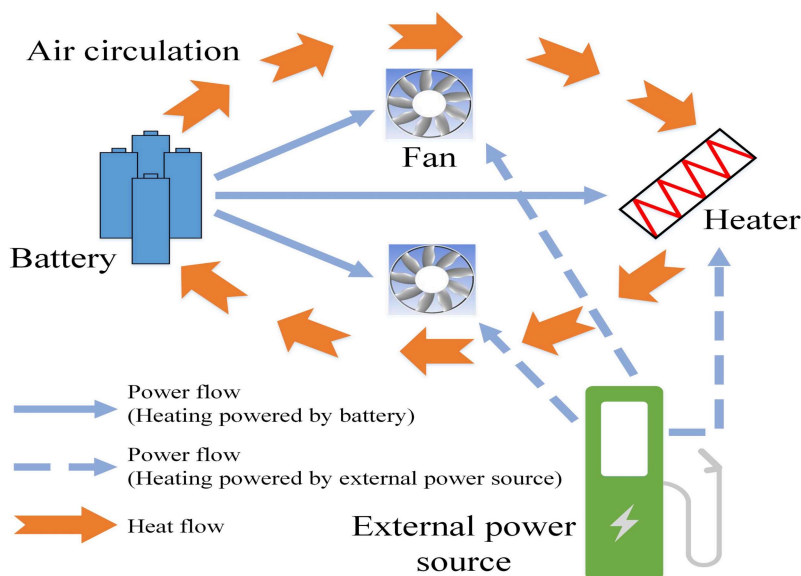


Fig. 5 Schematic of an air heating system. (Adapted from Ji Y, Wang CY. Heating strategies for Li-ion batteries operated from subzero temperatures. *Electrochimica Acta*. 2013; 107: 664-74.)

Internal heating

The internal heating strategy generates heat directly through the battery resistance, so the preheating system usually does not require too many control elements. Therefore, the heat transfer in such a system could be simple. The internal preheating method can usually heat the battery faster and more uniformly, and has certain advantages in terms of energy consumption, temperature uniformity, and cost (self-heating batteries as shown in Figure 6). However, because this type of preheating technology is not mature enough at the battery module/package level, it has not yet been commercialized. One of the main concerns for internal heating strategies lies in concomitant battery degradation. For internal self-heating and SHLB, since the battery discharges during heating, direct damage such as lithium plating does not happen, but high rate operations will aggravate aging in the long term. For the mutual pulse heating, using a low pulse frequency and heating at a high initial SOC should be circumvented since both of them increase the risk of lithium plating. For the AC heating, the current with a low frequency should be used cautiously with additional limitations (such as voltage bounds) to avoid irreversible damage to the battery. Using a high frequency in AC heating is conducive to reducing battery capacity loss.

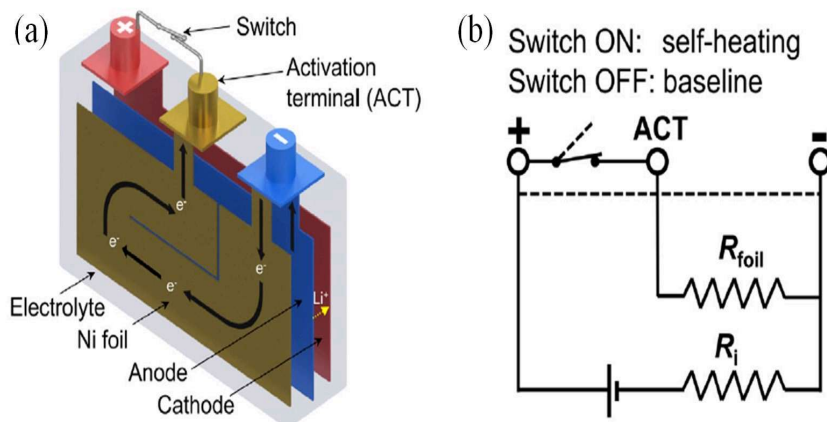


Fig. 6 Internal structure of the self-heating lithium-ion battery and its working principle (Adapted from Zhang G, Ge S, Xu T, Yang X-G, Tian H, Wang C-Y. Rapid self-heating and internal temperature sensing of lithium-ion batteries at low temperatures. *Electrochimica Acta*. 2016; 218: 149-55.)

Future trends

Despite the extensive research progress made so far, how to preheat batteries from subzero temperatures efficiently is still a challenge and remains to be unsolved. Thus the future trends are suggested from the following aspects to encourage more breakthroughs in this research area:

(1) The heat transfer process in the battery preheating system needs to be comprehensively designed from thermal science perspectives to enhance the heating performance, particularly in external heating. The heat transfer path and patterns are required to be optimized in order to reduce the heat transfer distance, enhance the heat transfer coefficient, and minimize the unwanted heat loss to the ambient environment. Therefore, better thermal insulation, adding more thermally conductive materials (e.g., nanofluids and CPCM), and applying more thermally efficient heating elements (e.g., heat pumps and film-based panel heaters) would also be beneficial to such preheating systems.

(2) The development of model-based control strategies in BTMS can achieve optimal battery heating for both external and internal heating. Current studies primarily focus on the design of preheating systems and the exploration of novel heating approaches. Although some inherent parameters of such heating systems have been investigated in terms of their effects on heating performance, the control optimality, which helps to achieve better heating performance through heating process controls, has seldom been studied. Nevertheless, model-based preheating control strategies typically require applicable models that are suitable for control and implementation. Therefore, the key challenge is to develop control-orientated battery preheating models with adequate accuracy as well as high computational efficiency, during which precise modeling of battery thermal behavior (at the cell, module, and pack level), system heat transfer characteristics, and system heat generation are necessary.

(3) The preheating strategies need to be further explored in a battery module/pack level since cell temperature homogeneity in a pack is critical to the overall performance of the battery pack and would affect its aging processes. If there is no heat generation inside the battery pack, namely the battery heating is a pure heat transfer problem, the effect of surrounding-to-cell heat interaction characteristic on the temperature uniformity of the pack can be studied. If heat is generated inside the battery pack, different cell heat generation amounts caused by cell resistance inconsistency could contribute to the thermal gradient inside the pack and, thus, need to be considered.

(4) The effect of preheating on battery aging remains to be investigated systematically, especially for AC heating. The effect of current parameters such as amplitude and frequency on battery aging needs to be studied from a perspective of electrochemical mechanisms inside a Li-ion battery. Electrochemical models governing the kinetic and transport processes at the anode under an AC signal load can be established to reveal the effects of AC parameters on aging explicitly. In this way, the ultimate current limits in AC heating could be fully explored to boost heating performance without harming the battery, making AC heating a favorable paradigm in battery low-temperature applications.

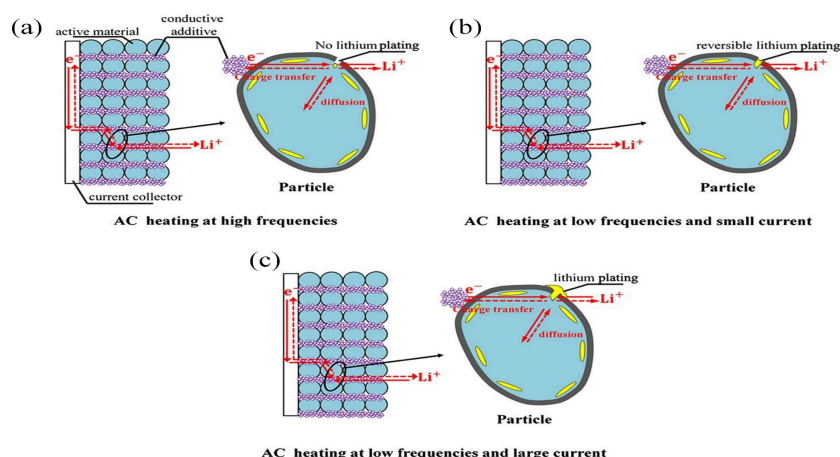


Fig. 7 Effect of AC heating on Li-ion battery aging with different current frequencies. (Adapted from Zhu J, Sun Z, Wei X, Dai H. An alternating current heating method for lithium-ion batteries from subzero temperatures. *International Journal of Energy Research*. 2016; 40: 1869-83.)

Publication information

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Extra introduction

Progress in Energy and Combustion Science

Progress in Energy and Combustion Science (PECS) is a top journal in the field of energy and combustion. It only invites world-renowned experts in the field of energy to write high-level review papers and does not accept free submissions. It publishes 6 issues per year, 4-6 articles per issue. The latest five-year impact factor is 26.747, ranking first in the fields of mechanical engineering (132 journals) and thermodynamics (58 journals).

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